TITLE: Interfacing of Stepper motor to 8051- software delay using Timer

AIM:

This experiment includes the interfacing of Stepper motor with 8051 microcontrollers. After completion of this experiment, students will be able to

I. Interface a 8051 microcontroller with a stepper motor.

II. Program a 8051 microcontroller using Embedded C Language.

THEORY:

Task 1: Stepper Motor interfacing with 8051 microcontrollers.

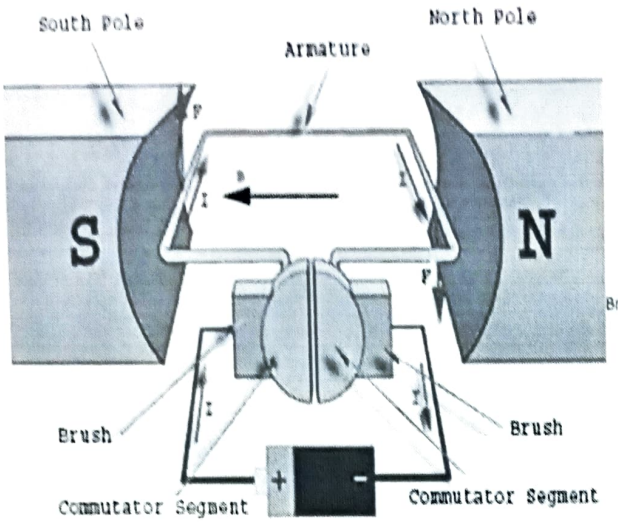
Theory:

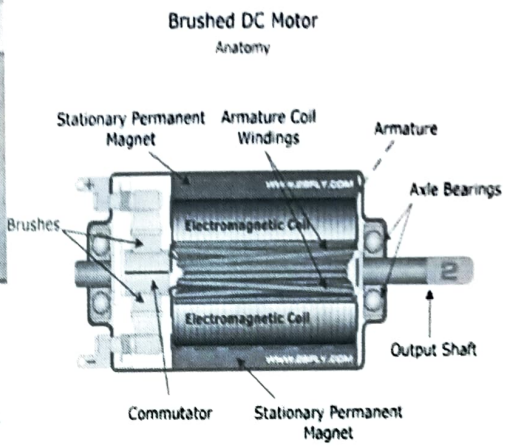


Stepper Motor

A DC motor is an electrical mechanism that converts direct current electrical power into mechanical power. The principle of working of a DC motor is that "whenever a current carrying conductor is placed in a magnetic field, it experiences a mechanical force".

The direction of this force is given by Fleming's left-hand rule.



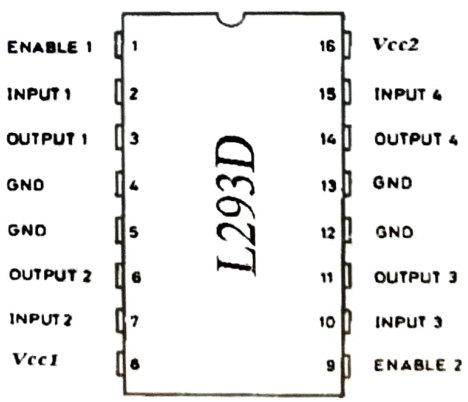


Hence by just reversing the voltage levels at the two pins of DC motor, we can reverse the direction of rotation of the shaft.

If two pins are at same levels i.e. 0v-0v or 12v-12v, the shaft will not move.

The need for the motor driver :

In practice 8051 microcontroller output pin sources low current not sufficient to drive the dc motor, also isolation is required in the interfacing of inductive devices. Hence, we normally use driver ICs such as ULN2003, L293D, transistors.



FUNCTION TABLE (each driver)

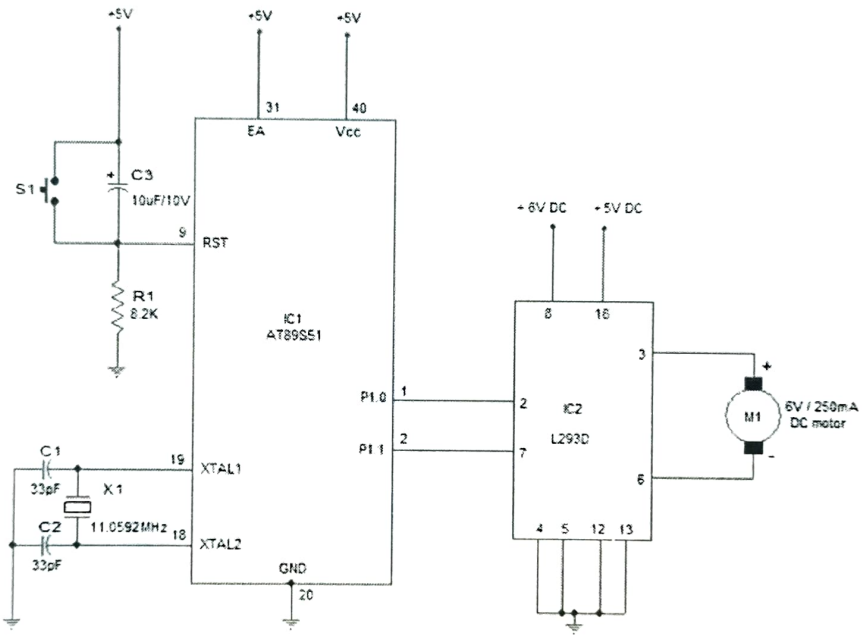
INPUTS		OUTPUT
A	EN	Y
H	H	H
L	H	L
X	L	Z

H = high level, L = low level,
X = irrelevant,
Z = high impedance (off)

Motor drivers are used preferably to

1. Increases voltage and current sourcing capability.
2. Provides isolation to 8051 from inductive effects of motors.
3. Motor drivers have inbuilt H bridges so as to reverse the direction of rotation, no rewiring is required.
4. En pin is provided for PWM speed control.

Interfacing diagram of Stepper motor with 8051 Microcontroller



PROGRAM:

CONCLUSION:

QUESTIONS:

1. What is the Difference between Conventional Motor and Stepper Motor.
2. List out Stepper Motor Advantages and Disadvantages.
3. Briefly describe the different types of Stepper Motor.

REFERENCES:

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[02] Datasheet: Available online.

[03] Mahumad Ali Mazadi, Janice Gillispie Mazidi, Rolind D. McKinlay, "8051 Microcontroller & Embedded Systems", Pearson Prentice Hall publication, IInd edition.